

VentureIQ — Viva Defense Questions & Answers

Essential Technical & Architectural Q&A; Guide

Q1: What is VentureIQ?

VentureIQ is an AI-powered startup due-diligence engine using LangGraph multi-agent workflows and Pinecone vector RAG to validate startup ideas, analyze markets, map competitors, and evaluate business model viability.

Q2: Why use multiple AI agents instead of a single prompt in ChatGPT?

A single prompt suffers from context pollution and superficiality. Specialized agents (Market, Competitor, Business, Risk) focus strictly on one domain with dedicated prompts, evidence context, and specialized tools.

Q3: What is LangGraph and why is it used?

LangGraph is a stateful orchestration framework that manages complex multi-agent workflows as directed state graphs, enabling state persistence, conditional task routing, and parallel execution.

Q4: What is RAG and why is it included?

RAG (Retrieval-Augmented Generation) retrieves relevant external documents from Pinecone before LLM generation, grounding agent analyses on empirical startup benchmark evidence rather than static training memory.

Q5: What are embeddings and how do they work in VentureIQ?

Embeddings are dense 768-dim numerical vectors where semantically similar text is close together. VentureIQ uses Google's models/text-embedding-004 to vectorize queries for similarity search against Pinecone.

Q6: What does Tavily do in your architecture?

Tavily is a real-time web search API built for AI agents. It searches live competitor pricing and market updates, complementing Pinecone's internal vector knowledge base.

Q7: What happens if Pinecone or network fails during the demo?

VentureIQ implements graceful fallbacks. If PINECONE_API_KEY is missing or Pinecone is offline, retrieve_context() logs a warning and sets retrieved_context="" without crashing the pipeline.

Q8: How are hallucinations reduced?

By grounding prompts on vector evidence, instructing agents to separate evidence from inference, using heuristic keyword checks, and failing gracefully when evidence is missing.